Review: Left Hand and Right Hand Sums

- If $f(x)$ is *decreasing* on $[a, b]$, then using a left-hand sum to estimate the area 'under' the graph of $f(x)$ between $x = a$ and $x = b$ gives an _____ of the true area, and using a right-hand sum gives an _____ of the true area.
- If $f(x)$ is *increasing* on $[a, b]$, then using a left-hand sum to estimate the area 'under' the graph of $f(x)$ between $x = a$ and $x = b$ gives an _____ of the true area, and using a right-hand sum gives an _____ of the true area.

Draw examples to show why the above are true.

To estimate the area, we divide $[a, b]$ into n equal subintervals of width $\Delta x =$ _____.

- In a left-hand sum, the area of the ith rectangle is _____.
- In a right-hand sum, the area of the ith rectangle is _____.

Question Consider the function $f(x) = (x-1)(x-2)$. Draw two copies of a graph of this function, then draw a left-hand sum and a right-hand sum with $n = 3$ subintervals for the area 'under' the graph between $x = 0$ and $x = 1.5$. Compute both sums.

Why did I put 'under' in quotes?

The Midpoint and Trapezoid Sums

Definition In a *midpoint sum*, we choose to compute the height of each rectangle in the *center* of each subinterval instead of on the left or right.

Question Fill in the blanks:

- When we divide the interval $[a, b]$ into n equal subintervals, the left hand of the ith subinterval is _____, and its right hand is _____.
- So the center of the ith subinterval is _____.
- The area of the ith rectangle in a midpoint sum is therefore _____.

Definition The *trapezoid sum* is the average of the left hand and right hand sums.

Definitions LHS_n refers to the left-hand sum with n subintervals. RHS_n refers to the right-hand sum with n subintervals. MPS_n refers to the midpoint sum with n subintervals. $TRAP_n$ refers to the Trapezoid sum with n subintervals.

For Questions 1 and 2 below, draw and then calculate $(LHS)_2$, $(RHS)_2$, $(TRAP)_2$ and $(MPS)_2$ for the given integrals:

1. $\int_0^4 x^2 - 1 \, dx$.

□Regrade

$(LHS)_2 =$ ___ $(RHS)_2 =$ ____ $(TRAP)_2 =$ ____ $(MPS)_2 =$ ____

2. $\int_0^4 \sqrt{x} - 2\, dx.$
□Regrade

$(LHS)_2 \approx$ _____ $(RHS)_2 \approx$ _____ $(TRAP)_2 \approx$ _____ $(MPS)_2 \approx$ _____

3. (a) In both Question 1 and in Question 2, _____ is an overestimate of the integral, and _____ is an underestimate. This is true because both functions are _____.
□Regrade

(b) Assuming that in each case one of $TRAP$ and MPS is an overestimate of the true area, and the other an underestimate, which one is which in each of the two cases above?

(c) What difference in shape between the two functions that might explain this?

Why Does Concavity Matter?

Let's demonstrate why your answer to the last question is true. To do this, we must recast these two sums as sum of areas of *trapezoids*:

Conclusion

4. □Regrade

- If $f(x)$ is _____, then LHS_n is an underestimate of the true 'area' under $f(x)$ and RHS_n is an overestimate for it.
- If $f(x)$ is _____, then LHS_n is an overestimate of the true 'area' under $f(x)$ and RHS_n is an underestimate for it.
- If $f(x)$ is _____, then MPS_n is an underestimate of the true 'area' under $f(x)$ and $TRAP_n = \frac{LHS_2+RHS_2}{2}$ is an overestimate for it.
- If $f(x)$ is _____, then MPS_n is an overestimate of the true 'area' under $f(x)$ and $TRAP_n = \frac{LHS_2+RHS_2}{2}$ is an underestimate for it.